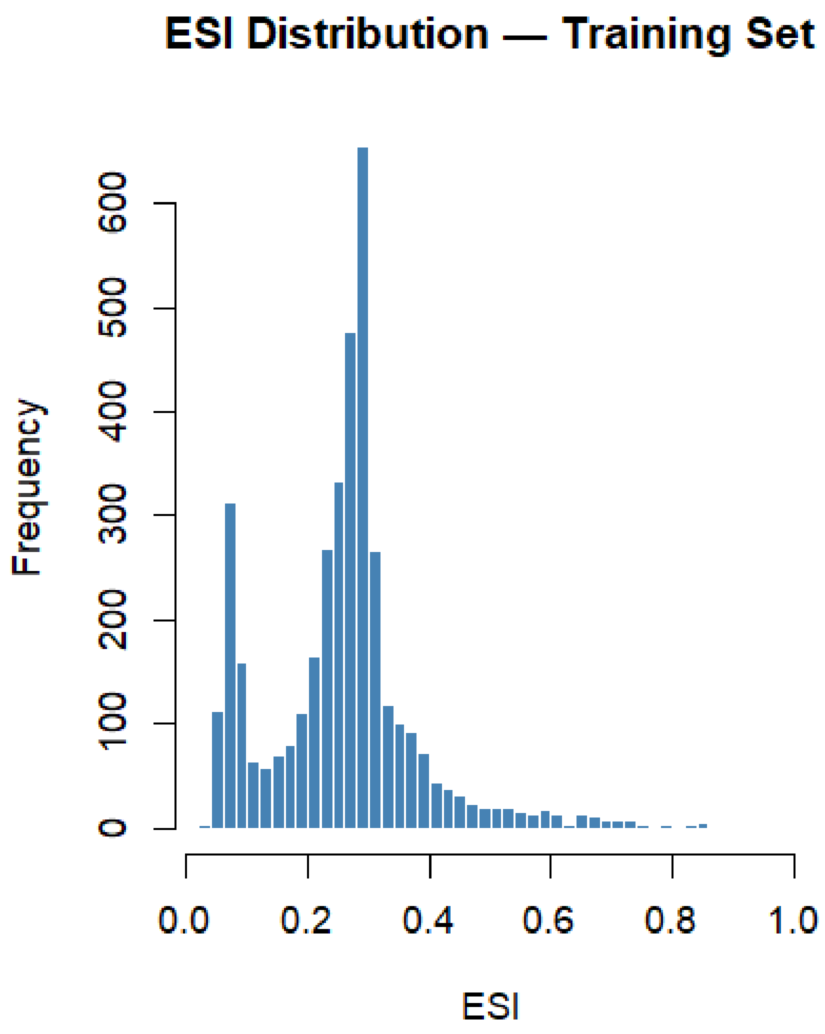
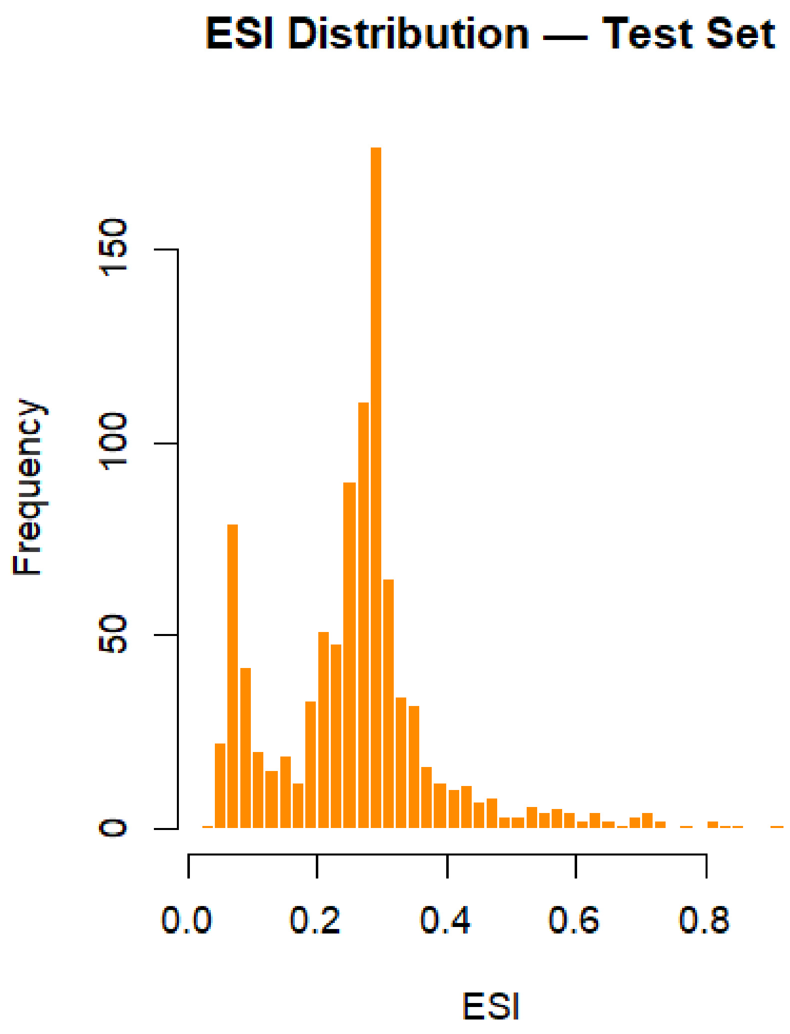
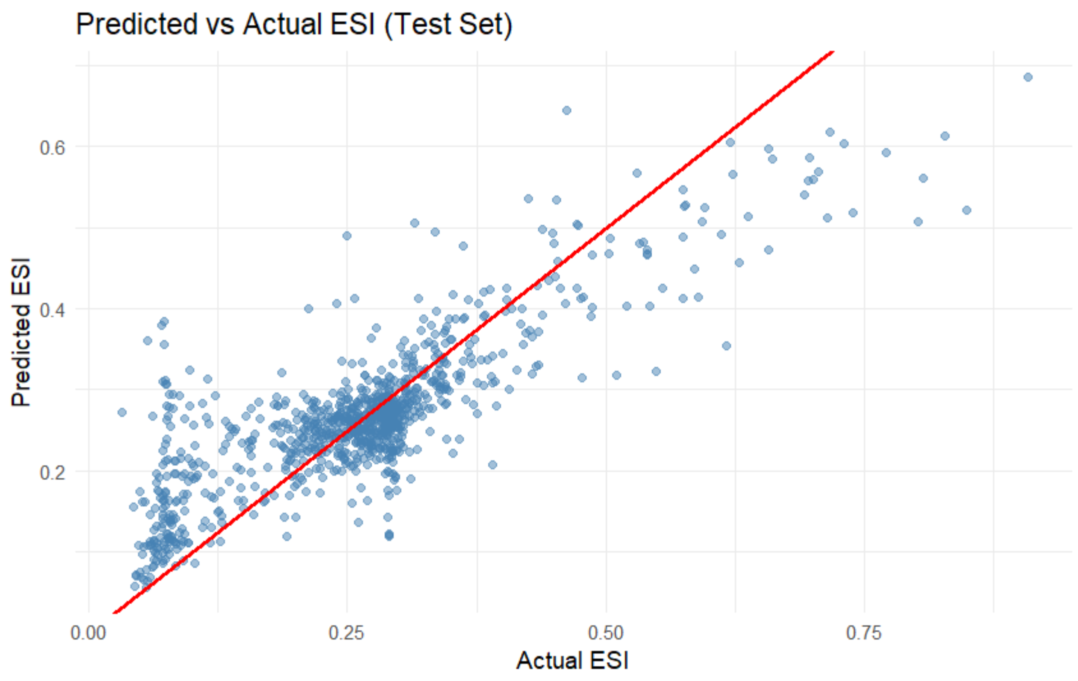
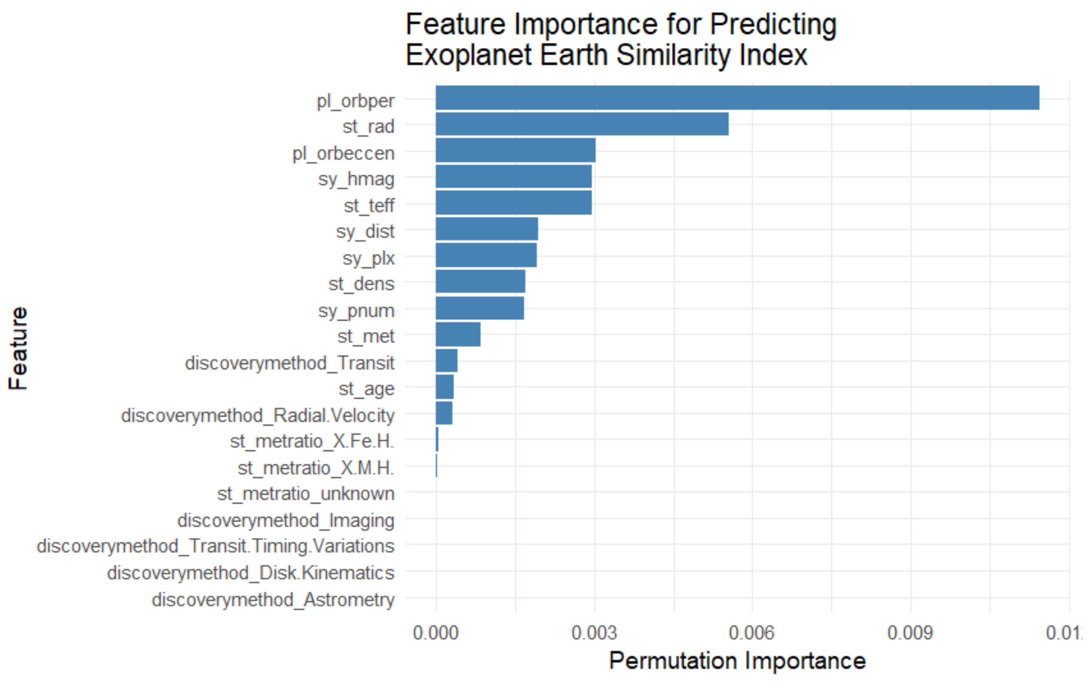
# Can We Predict How Earth-Like a Planet Is From Its Star and Orbit Alone?

- **Goal:** Predict the Earth Similarity Index (ESI) of confirmed exoplanets using only stellar and orbital features, without any knowledge of a planet's physical composition
- **Data:**
  - NASA Exoplanet Archive: Planetary Systems Composite Data
  - Planetary Habitability Laboratory (PHL) @ UPR Arecibo: Habitable Worlds Catalog (HWC) Data
- **Response variate** is ESI
  - A continuous score where 0 means unlike Earth and 1 means identical to Earth
- **Features:** Orbital features (e.g. orbital period), stellar features (e.g. stellar temperature), and system-level features (e.g. distance to system)
  - Planetary physical composition features were explicitly excluded
- **Model:** Random Forest regression
  - Able to handle non-linear relationships and produce interpretable feature importances


ESI Distribution — Training Set
Frequency
0
100
200
300
400
500
600
0.0
0.2
0.4
0.6
0.8
1.0
ESI


ESI Distribution — Test Set
Frequency
0
50
100
150
0.0
0.2
0.4
0.6
0.8
ESI


Predicted vs Actual ESI (Test Set)
0.6
0.4
0.2
Predicted ESI
0.00
0.25
0.50
0.75
Actual ESI


Feature Importance for Predicting
Exoplanet Earth Similarity Index
pl_orbper
st_rad
pl_orbeccen
sy_hmag
st_teff
sy_dist
sy_plx
st_dens
sy_pnum
st_met
discoverymethod_Transit
st_age
discoverymethod_Radial.Velocity
st_metratio_X.Fe.H.
st_metratio_X.M.H.
st_metratio_unknown
discoverymethod_Imaging
discoverymethod_Transit.Timing.Variations
discoverymethod_Disk.Kinematics
discoverymethod_Astrometry
Feature
0.000
0.003
0.006
0.009
Permutation Importance